Safe Consensus Tracking With Guaranteed Full State and Input Constraints: A Control Barrier Function-Based Approach

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***Abstract*—In this article, we consider the safe consensus tracking problem for uncertain second-order nonlinear multiagent systems subject to position, velocity, and input constraints. The agents are required to cooperatively track a desired leader's trajectory while always satisfying their local state and input constraints. Therefore, conflicting objectives may exist for an agent when the desired trajectory violates its local constraints. We propose to solve this problem using a control barrier function (CBF)-based approach. The cooperative tracking objective is encoded by a novel control Lyapunov function-based condition while the state and input constraints are handled by CBF-based constraints. For the relative degree two position constraint, two classes of CBF-based conditions are proposed based on high-order CBFs and a modified CBF design, respectively. It is proven that, with the modified CBF, there always exist feasible control inputs that satisfy all the CBF-based constraints. Then, unified quadratic programming-based controllers are formulated and the performances are analyzed. Simulation examples are provided to verify the obtained results.**

***Index Terms*—Control barrier function, input saturation, multiagent system, safe consensus tracking, state constraint.**

I. Introduction

Due to the rapid development of sensing, communication, and computation technologies, multiagent system has seen increasing use in various fields, such as environmental sensing, cooperative transportation, target tracking, and so on [1]. Among the various cooperative control tasks of the multiagent systems, consensus tracking plays a fundamental role as many other tasks, such as formation, containment, and encirclement, can be solved based on the consensus tracking controllers [2]. Many important results have been obtained for consensus tracking of multiagent systems handling different agent dynamics, communication graphs, and convergence speed requirement [3], [4].

It should be noted that most of the existing results have focused on the consensus tracking of multiagent systems without considering the physical constraints of the agents in practical applications. Since all physical plants can only have limited actuation power, input saturation effect should be taken into consideration in order to guarantee the stability of the closed-loop system. In [5] and [6], global nonlinear consensus and formation tracking controllers have been proposed for agents with simple dynamics. Low-gain-based method has been employed in [7] and [8] to achieve global or semiglobal consensus of linear multiagent systems. Model predictive control (MPC) method has been generalized to handle consensus and formation control of nonlinear multiagent systems in[9] and [10].

Apart from the input saturation, the state of the agents may also have constraints due to performance or safety concerns. For example, for multiple fixed-wing aircraft, the velocities of the agents should have lower bound in order to keep flying and for multiple autonomous vehicles, the velocities usually have upper bound due to collision avoidance requirements. For multiple manipulators in the working space, both the positions and velocities of the agents may be constrained due to the safety operation protocols. In these situations, all the state and input constraints should be simultaneously taken into consideration when designing the consensus tracking controllers. Several methodologies have been proposed to handle these constraints in existing literature. The first class of methods aims to directly propose nonlinear distributed controllers such that both the state and input constraints can be satisfied. In [11] and [12], nonlinear distributed controllers have been proposed, which achieve consensus tracking of second-order integrator systems with both the velocity and input constraints. In [13], a control barrier function (CBF)-based method was employed to achieve collision avoidance between the agents and the obstacles. However, these results were limited to integrator systems without uncertainties, and the position constraints were not considered simultaneously. The second class of methods generalizes the barrier Lyaupnov function (BLF) method to multiagent systems, which could solve the full state and input-constrained consensus problems [14], [15]. Moreover, to relax the feasibility conditions, nonlinear state transformation-based methods have been given in [16] and [17] to achieve constrained consensus tracking of multiagent systems. Another class of methods for state and input constrained cooperative control of multiagent systems is the MPC-based controllers [18], [19], which can handle nonlinear dynamics and constraints better. Note that even though most of the existing MPC-based results have not considered the position constraints, they can be incorporated into the local optimization problem formulation. However, the MPC-based methods generally require strict initial feasibility conditions to maintain the feasibility of the optimization problems and an accurate model of the agent dynamics is usually needed to guarantee the performance. Moreover, the computation costs may hinder real-time implementation on practical multiagent systems.

Although there have been some initial progress in addressing the state and input constrained consensus tracking of multiagent systems as abovementioned, a common limitation of these results is that the desired trajectory has to lie in the constrained region of all the follower agents. This condition is central for the construction of Lyapunov functions

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in the BLF-based and nonlinear state transformation-based methods, and for the stability analysis in the MPC-based methods. However, since the leader's trajectory may be given by an external agent or human operator, it may easily be the case that the desired trajectory violates the constraints of the follower agents occasionally. In these cases, a more desired behavior of the multiagent system should be safe consensus tracking, which means the agents should always satisfy their local constraints while achieving conventional consensus tracking only when the desired trajectory is feasible for all the agents in the network. However, to the best of authors' knowledge, an effective control strategy that achieves this objective is still lacking.

Inspired by the control Lyapunov function (CLF)-CBF-based control framework proposed in [20] for single systems, we propose a novel CBF-based safe consensus tracking strategy for second-order uncertain nonlinear multiagent systems. The consensus tracking objective is encoded by a novel CLF-based condition relying on only local relative state information between neighboring agents. The state and input constraints of the agents are handled by several CBFs, which provide the constraints on the local control inputs at each time instant. Finally, quadratic programming (QP)-based controllers are proposed to achieve the safe consensus tracking control objective. The main contributions of this work lie in the following aspects. First, we formulate a safe consensus tracking problem for second-order uncertain nonlinear multiagent systems where the leader's trajectory is not required to lie in the constrained region of all the followers. Second, a CBF-based approach is proposed, which solves the safe consensus tracking problem successfully. A new type of CBF is then proposed, which guarantees the feasibility of local QP problems. Third, the proposed method does not need extensive computation as in the MPC-based methods and, therefore, is more suitable for practical applications.

The rest of this article is organized as follows. In Section II, some preliminaries and the problem formulation are given. In Section III, the CBF-based safe consensus tracking control strategy is presented. Simulation examples are given in Section IV. Finally, Section V concludes this article.

II. Preliminaries and Problem Formulation

A. Preliminaries

In this section, some preliminaries on the CLF-CBF-based control method for single systems is presented. Consider a nonlinear affine control system

$$\dot{x} = f(x) + g(x)u \tag{1}$$

where f and g are locally Lipschitz, and $x \in D \subset \mathbb{R}^n$ and $u \in U \subset \mathbb{R}^m$ are the set of admissible inputs. V is a CLF for (1) if it is positive definite and satisfies $\inf_{u\in U}[L_f V(x) + L_g V(x)u] \leqslant -\gamma(V(x))$, where γ is a class $\mathcal{K}$ function. Consider the set $K_{\text{clf}}(x) = \{u \in U : L_f V(x) + L_g V(x)u \leqslant -\gamma(V(x))\}$ for every point $x \in D$. The following lemma can be derived from [21].

Lemma 1: For the nonlinear control system (1), if there exists a CLF $V : D \to \mathbb{R}_{\geq 0}$, then any locally bounded and measurable feedback controller $u(x) \in K_{\text{clf}}(x)$ asymptotically stabilizes the system to $x^* = 0$.

Safety of (1) can be framed in the context of enforcing invariance of a set. Consider a set $\mathcal{C}$ defined as the superlevel set of a continuously differentiable function $h : D \subset \mathbb{R}^n \to \mathbb{R}$ with $\mathcal{C} = \{x \in D \subset \mathbb{R}^n : h(x) \geq 0)\}$. The system (1) is said to be safe with respect to the set $\mathcal{C}$ if the set $\mathcal{C}$ is forward invariant. The function h is a CBF if there exists an extended class $\mathcal{K}_\infty$ function α such that it holds $\sup_{u\in U}[L_f h(x) + L_g h(x)u] \geqslant -\alpha(h(x))$ for all $x \in D$ [20]. Consider the set $K_{\text{cbf}}(x) = \{u \in U : L_f h(x) + L_g h(x)u + \alpha(h(x)) \geqslant 0\}$.

Lemma 2 ([22]): For the nonlinear control system (1), if h is a CBF on D and $\frac{\partial h}{\partial x}(x) \neq 0$ for all $x \in \partial\mathcal{C}$, then any locally bounded and measurable controller $u(x) \in K_{\text{cbf}}(x)$ renders the set $\mathcal{C}$ safe.

Based on the CLF and CBF, the following QP-based controller can be formulated to unify stability and safety:

$$\begin{aligned} u(x) = &\mathop{\arg\min}_{(u,\delta)\in\mathbb{R}^{m+1}} \tfrac{1}{2}u^T H(x)u + p\delta^2 \\ \text{s.t.}\ \ &L_f V(x) + L_g V(x)u + \gamma(V(x)) \leqslant \delta \\ &L_f h(x) + L_g h(x)u + \alpha(h(x)) \geqslant 0 \end{aligned}$$

where $H(x)$ is any positive definite matrix and δ is a relaxation variable penalized by $p > 0$, which relaxes the condition on stability to guarantee safety.

B. Problem Formulation

Consider a second-order nonlinear multiagent system

$$\begin{aligned} \dot{x}_{i1} &= x_{i2} \\ \dot{x}_{i2} &= f_i(x_i) + g_i(x_i)u_i, i = 1, \ldots, N \end{aligned} \tag{2}$$

where $x_i = [x_{i1}, x_{i2}]^T \in \mathbb{R}^2$ is the state of agent i, $u_i \in \mathbb{R}$ is the control input, $f_i(x_i) : \mathbb{R}^2 \to \mathbb{R}$ is the nonlinear dynamics, and $g_i(x_i) : \mathbb{R}^2 \to \mathbb{R}$ is the control gain. The safe consensus tracking problem of (2) is considered, and the leader's trajectory is given by a differentiable function $x_{01}(t)$ with $|\dot{x}_{01}(t)| \leq v_L$, where $v_L \geq 0$. For agent i, suppose that the position and velocity constraints $|x_{i1}| \leq p_i$ and $|x_{i2}| \leq v_i$ are required for performance or safety concerns. The input constraint $|u_i| \leq q_i$ is required due to the actuator saturation constraint. For the agent dynamics, suppose that the nonlinear dynamics f_i is unknown and only an approximate $\hat{f}_i$ is known, which satisfies $|f_i(x_i) - \hat{f}_i(x_i)| \leq \alpha_i$ and $|\hat{f}_i(x_i)| \leq \beta_i$ with two positive constants $\alpha_i, \beta_i > 0$ on the region $\{x_i : |x_{i1}| \leq p_i, |x_{i2}| \leq v_i\}$. The control gain $g_i(x_i) > 0$ is assumed to be known and has the lower and upper bounds of $\underline{g}_i$ and $\bar{g}_i$, respectively.

Remark 1: The assumption of known gain function $g_i(x_i)$ is not restrictive since if only an approximate $\hat{g}_i$ is available, which satisfies $|\hat{g}_i - g_i| \leq \bar{\alpha}_i$, then by redefining f_i as $f_i + (g_i - \hat{g}_i)u_i$, we can transform the agent dynamics into the form that we consider. Furthermore, the obtained results can be easily generalized to the cases $x_i \in \mathbb{R}^{2n}$ where $n > 1$.

With the input transformation $u_i = \frac{1}{g_i}w_i$, we have $\dot{x}_{i2} = w_i + f_i$. Moreover, when $|w_i| \leq q_i\underline{g}_i$, we have $|u_i| \leq q_i$.

Let the interaction relation among the N agents be described by an undirected and connected graph $\mathcal{G}$. The adjacency matrix $A := [a_{ij}] \in \mathbb{R}^{N\times N}$ satisfies $a_{ij} > 0$ if agents i and j can obtain information from each other and $a_{ij} = 0$, otherwise. The Laplacian matrix $L := [l_{ij}] \in \mathbb{R}^{N\times N}$ is defined by $l_{ii} = \sum_{j=1}^N a_{ij}$ and $l_{ij} = -a_{ij}$ for $i \neq j$. Moreover, $a_{i0} > 0$ if agent i can directly obtain the information of the leader and $a_{i0} = 0$, otherwise. Let $A_0 = \text{diag}\{a_{10}, \ldots, a_{N0}\}$. Then, the matrix $H = L + A_0$ is positive definite under the condition that at least one agent can directly obtain the information of the leader [7].

In the conventional consensus tracking problem, the control objective is usually to design the control input u_i for each agent such that the states of all the agents gradually approach the common trajectory of the leader, i.e., $x_{i1}(t) - x_{01}(t) \to 0$ as $t \to \infty$. When the state and input constraints of the agents are presented, this consensus tracking requirement may be in conflict with the local constraints of the agents. In these cases, the safe consensus tracking objective is preferred, which means when the leader's trajectory is not in conflict with the constraints of the agents, consensus tracking can be achieved. When conflict exists, the constraints of the agents are prioritized and compromised consensus tracking performance is tolerated. As a result, a multiobjective control problem is considered in this article, and we solve this problem using a CBF-based approach.

III. CBF-Based Safe Consensus Tracking Control

A. CLF-Based Consensus Tracking Condition

In order to transform the distributed consensus tracking problem into a form suitable for the CLF-based method, we first consider the following desired dynamics for the multiagent system:

$$\dot{x}_{i1} = -k_1 \tanh(\xi_i) - k_2 \tanh(\xi_i/\varepsilon) \tag{3}$$

where $\xi_i = \sum_{j=0}^{N} a_{ij}(x_{i1} - x_{j1})$, $k_1 > 0$, $k_2 \geq v_L$, and $\varepsilon > 0$ is a small positive parameter.

Remark 2: The desired dynamics (3) is motivated by the distributed consensus tracking controller for first-order multiagent systems. It will be used to construct the CLFs in the later development. The first hyperbolic tangent function can be replaced by any globally bounded strictly increasing odd function while the second term is a continuous approximation of the sign function with the small positive $\varepsilon > 0$.

We show that under the desired dynamics (3), practical consensus tracking of the multiagent system (2) can be achieved.

Theorem 1: Suppose the interaction graph $\mathcal{G}$ is undirected and connected and at least one follower can directly obtain the information of the leader. Along the desired dynamics (3), the consensus tracking errors $\tilde{x}_{i1} = x_{i1} - x_{01}$, $i = 1, \ldots, N$ are uniformly ultimately bounded by a positive number, which approaches zero as ε approaches zero.

Proof: Let $\tilde{x} = [\tilde{x}_{11}, \ldots, \tilde{x}_{N1}]^T$ and $\xi = [\xi_1, \ldots, \xi_N]^T$. We have that $\xi = H\tilde{x}$ and $\|\xi\|_2^2 = \tilde{x}^T H^2 \tilde{x} \geqslant \underline{\lambda}^2 \|\tilde{x}\|_2^2$, where $\underline{\lambda} = \lambda_{\min}(H) > 0$. From (3), we have

$$\dot{\tilde{x}} = -k_1 \tanh(\xi) - k_2 \tanh(\xi/\varepsilon) - 1_N \dot{x}_{01} \tag{4}$$

where $1_N \in \mathbb{R}^N$ is the vector of all ones. Consider the Lyapunov function $V = \frac{1}{2}\tilde{x}^T H \tilde{x}$. It follows that

$$\dot{V} = -k_1 \xi^T \tanh(\xi) - k_2 \xi^T \tanh(\xi/\varepsilon) - \xi^T 1_N \dot{x}_{01}$$
$$\leqslant -k_1 \xi^T \tanh(\xi) - \sum_{i=1}^{N} [k_2 \xi_i \tanh(\xi_i/\varepsilon) - v_L |\xi_i|].$$

From [23, Lemma 1], we have that for any $\eta \in \mathbb{R}$ and $\varepsilon > 0$, it holds $0 \leq |\eta| - \eta \tanh(\eta/\varepsilon) \leq \kappa\varepsilon$, where $\kappa = 0.2785$. Then, it follows:

$$\dot{V} \leqslant -k_1 \xi^T \tanh(\xi) + k_2 N \kappa \varepsilon$$

where we have used the condition $k_2 \geq v_L$. Let $c > 0$ be a constant, which satisfies $c \tanh(c) > \frac{k_2 N \kappa \varepsilon}{k_1}$. It is easy to see that c can be taken arbitrarily small as ε approaches zero. Under the condition $V \geqslant \frac{\bar{\lambda}}{2\underline{\lambda}^2} N c^2$, noting that $V \leqslant \frac{1}{2}\bar{\lambda}\|\tilde{x}\|_2^2$ where $\bar{\lambda} = \lambda_{\max}(H) > 0$, we have $\|\xi\|_2^2 \geqslant \underline{\lambda}^2 \|\tilde{x}\|_2^2 \geqslant N c^2$, which leads to $\|\xi\|_\infty \geqslant \frac{\|\xi\|_2}{\sqrt{N}} \geqslant c$. Then, it follows that $\dot{V} \leqslant -k_1 c \tanh(c) + k_2 N \kappa \varepsilon < 0$.

Therefore, the set $V \leqslant \frac{\bar{\lambda}}{2\underline{\lambda}^2} N c^2$ is globally attractive and positive invariant. Noting that $\frac{1}{2}\underline{\lambda}\|\tilde{x}\|_2^2 \leqslant V$, it follows that $\|\tilde{x}\|_2$ is uniformly ultimately bounded by $\sqrt{N\bar{\lambda}/\underline{\lambda}^3}c$, which leads to that $\tilde{x}_{i1}, i = 1, \ldots, N$ are uniformly ultimately bounded by $\sqrt{N\bar{\lambda}/\underline{\lambda}^3}c$. It is easy to see that the consensus tracking error bound approaches zero as ε approaches zero. □

The desired velocity (3) has the nice feature that it is globally bounded by $k_1 + k_2$. Therefore, it is possible to achieve this velocity while satisfying the velocity constraint by tuning the controller gains. Furthermore, the desired velocity only depends on the relative position information from neighbors. This fact can be utilized to obtain a local CLF-based consensus tracking condition for each agent. Specifically, we define the following Lyapunov function for the consensus tracking objective:

$$V_f = \sum_{i=1}^{N} [x_{i2} + k_1 \tanh(\xi_i) + k_2 \tanh(\xi_i/\varepsilon)]^2.$$

Note that when V_f converges to zero exponentially, we have $x_{i2} = x_{i2}^d + e_{i2}$, where $x_{i2}^d = -k_1 \tanh(\xi_i) - k_2 \tanh(\xi_i/\varepsilon)$ and e_{i2} is a term, which converges to zero exponentially. Following similar analysis as in the proof of Theorem 1, one can show that practical consensus tracking for the agents can be achieved. From Section II-A, the condition

$$\dot{V}_f + \lambda V_f \leq 0 \tag{5}$$

is imposed to achieve exponential convergence of the Lyapunov function V_f where $\lambda > 0$. Note that one can obtain a decoupled sufficient condition for (5) with

$$\dot{V}_{if} + \lambda V_{if} \leq 0 \tag{6}$$

for $i = 1, \ldots, N$, where $V_{if} = [x_{i2} + k_1 \tanh(\xi_i) + k_2 \tanh(\xi_i/\varepsilon)]^2$. Based on (3) and (6), we have the CLF-based consensus tracking condition $2(x_{i2} - x_{i2}^d)[w_i + f_i - \dot{x}_{i2}^d] + \lambda(x_{i2} - x_{i2}^d)^2 \leq 0$. Noting that f_i is unknown, the following robust CLF-based condition on w_i is employed:

$$2(x_{i2} - x_{i2}^d)w_i + 2\alpha_i \left|x_{i2} - x_{i2}^d\right|$$
$$+ 2(x_{i2} - x_{i2}^d)(\hat{f}_i - \dot{x}_{i2}^d) + \lambda(x_{i2} - x_{i2}^d)^2 \leq 0 \tag{7}$$

which is distributed since only local velocity and relative position, and velocity information with respect to neighbors are needed.

Remark 3: The CLF-based condition (7) can be generalized to handle other types of coordination tasks, such as distributed containment by employing different local desired dynamics (3). Moreover, general directed communication graphs may also be handled by showing the convergence of the desired dynamics with graph related Lyapunov functions.

B. CBF-Based Constraints Handling

1) Velocity Constraint: The velocity constraint is to be handled by the CBF candidate $h_{si} = v_i^2 - x_{i2}^2$. Note that h_{si} has relative degree one with respect to the agent dynamics (2). Therefore, the CBF-based condition $\dot{h}_{si} + \rho_1 h_{si} \geq 0$ can be used to obtain the local input constraint, which leads to $-2x_{i2}(f_i + w_i) + \rho_1(v_i^2 - x_{i2}^2) \geqslant 0$. Since f_i is unknown, the following robust condition on w_i is considered:

$$-2x_{i2}w_i - 2x_{i2}\hat{f}_i - 2\alpha_i |x_{i2}| + \rho_1 \left(v_i^2 - x_{i2}^2\right) \geqslant 0. \tag{8}$$

Note that under the condition $\alpha_i + \beta_i < q_i \underline{g}_i$, one can always choose the control input $w_i = -q_i \underline{g}_i \mathrm{sgn}(x_{i2})$ such that (8) is satisfied. That is, the CBF-based condition (8) is compatible with the input constraint

$$|w_i| \leq q_i \underline{g}_i. \tag{9}$$

2) Position Constraint: For the position constraint, one can similarly consider the CBF candidate $h_{ri} = p_i^2 - x_{i1}^2$. Note, however, that h_{ri} has relative degree two with respect to the agent dynamics (2). Therefore, high-order CBF should be utilized to obtain the constraint on the control input. Inspired by the exponential high-order CBF given in [24], the CBF-based condition $\ddot{h}_{ri} + (\rho_2 + \rho_3)\dot{h}_{ri} + \rho_2\rho_3 h_{ri} \geqslant 0$ is considered, where $\rho_2, \rho_3 > 0$. It follows that $-2x_{i1}(w_i + f_i) - 2x_{i2}^2 - 2x_{i1}x_{i2}(\rho_2 + \rho_3) + \rho_2\rho_3(p_i^2 - x_{i1}^2) \geqslant 0$. Noting that f_i is unknown, the following robust condition on w_i is derived:

$$\begin{aligned} &- 2x_{i1}w_i - 2x_{i1}\hat{f}_i - 2\alpha_i\left|x_{i1}\right| - 2x_{i2}^2 \\ &- 2x_{i1}x_{i2}\left(\rho_2 + \rho_3\right) + \rho_2\rho_3\left(p_i^2 - x_{i1}^2\right) \geqslant 0. \end{aligned} \tag{10}$$

Under (10) and the initial conditions $\dot{h}_{ri}(0) + \rho_2 h_{ri}(0) > 0$ and $h_{ri}(0) > 0$, the position constraint can be enforced.

Note that for the CBF-based condition (10), one cannot guarantee that it is compatible with both the velocity constraint induced condition (8) and the input constraint (9). When there are conflicts, there exists no control input, which satisfies all the constraints simultaneously and the safety of the agents cannot be guaranteed. Therefore, we consider an alternative position constraint CBF design, which guarantees feasibility in the presence of all the position, velocity, and input constraints. For agent i, with the initial position $x_{i1} \in [-p_i, p_i]$ and the initial velocity $x_{i2} \in [-v_i, v_i]$, when $x_{i2} > 0$, under the condition $\alpha_i + \beta_i < q_i\underline{g}_i$, one can guarantee a deceleration of at least $\gamma_i = q_i\underline{g}_i - \alpha_i - \beta_i$ with $w_i = -q_i\underline{g}_i\mathrm{sgn}(x_{i2})$ until the agent stops. Then, under the condition $p_i - x_{i1} - \frac{x_{i2}^2}{2\gamma_i} \geqslant 0$, the position constraint will be satisfied. When $x_{i2} < 0$, with the similar control strategy, under the condition $x_{i1} + p_i - \frac{x_{i2}^2}{2\gamma_i} \geqslant 0$, the position constraint can be guaranteed. Therefore, the following two CBF candidates are considered:

$$h_{ri}^{+}\left(x_{i1}, x_{i2}\right) = \begin{cases} p_i - x_{i1} - \frac{x_{i2}^2}{2\gamma_i}, x_{i2} \geqslant 0 \\ p_i - x_{i1}, x_{i2} < 0 \end{cases}$$

and

$$h_{ri}^{-}\left(x_{i1}, x_{i2}\right) = \begin{cases} p_i + x_{i1} - \frac{x_{i2}^2}{2\gamma_i}, x_{i2} \leqslant 0 \\ p_i + x_{i1}, x_{i2} > 0. \end{cases}$$

Both h_{ri}^{+} and h_{ri}^{-} are continuously differentiable functions with respect to x_{i1} and x_{i2}. Furthermore, when both $h_{ri}^{+} > 0$ and $h_{ri}^{-} > 0$ are maintained, we have $-p_i < x_{i1} < p_i$, i.e., the position constraint is satisfied. Moreover, both h_{ri}^{+} and h_{ri}^{-} have relative degree one with respect to the control input w_i.

The CBF-based conditions corresponding to h_{ri}^{+} and h_{ri}^{-} are $\dot{h}_{ri}^{+} + \rho_4 h_{ri}^{+} \geqslant 0$ and $\dot{h}_{ri}^{-} + \rho_4 h_{ri}^{-} \geqslant 0$, respectively. We can show that at any time instant, at most, one of the conditions is active. For example, when $x_{i2} > 0$, the CBF-based condition $\dot{h}_{ri}^{+} + \rho_4 h_{ri}^{+} \geqslant 0$ leads to $-\frac{x_{i2}}{\gamma_i}(w_i + f_i) - x_{i2} + \rho_4(p_i - x_{i1} - \frac{x_{i2}^2}{2\gamma_i}) \geqslant 0$. Since f_i is unknown, the following sufficient condition is considered:

$$-\frac{x_{i2}}{\gamma_i}w_i - \frac{x_{i2}}{\gamma_i}\hat{f}_i - \alpha_i\frac{|x_{i2}|}{\gamma_i} - x_{i2} + \rho_4\left(p_i - x_{i1} - \frac{x_{i2}^2}{2\gamma_i}\right) \geqslant 0. \tag{11}$$

For h_{ri}^{-}, the CBF-based condition is automatically satisfied. Similarly, when $x_{i2} < 0$, only the CBF-based condition

$$-\frac{x_{i2}}{\gamma_i}w_i - \frac{x_{i2}}{\gamma_i}\hat{f}_i - \alpha_i\frac{|x_{i2}|}{\gamma_i} + x_{i2} + \rho_4\left(p_i + x_{i1} - \frac{x_{i2}^2}{2\gamma_i}\right) \geqslant 0 \tag{12}$$

is active while $\dot{h}_{ri}^{+} + \rho_4 h_{ri}^{+} \geqslant 0$ is automatically satisfied. Moreover, when $x_{i2} = 0$, both the CBF-based conditions are automatically satisfied.

A crucial feature of the proposed CBFs h_{pi}^{+} and h_{pi}^{-} for the position constraint is that feasible control input, which satisfies all the position, velocity, and input constraints induced conditions, is guaranteed.

Theorem 2: Under the conditions $h_{ri}^{+}(0) > 0$, $h_{ri}^{-}(0) > 0$, and $h_{si}(0) > 0$, there always exists feasible control input $w_i \in [-q_i\underline{g}_i, q_i\underline{g}_i]$ such that $h_{ri}^{+}(t) > 0$, $h_{ri}^{-}(t) > 0$, and $h_{si}(t) > 0$ hold for $\forall t \geq 0$.

Proof: We prove the conclusion by showing that there always exists control input w_i such that the CBF-based conditions $\dot{h}_{ri}^{+} + \rho_4 h_{ri}^{+} \geqslant 0$, $\dot{h}_{ri}^{-} + \rho_4 h_{ri}^{-} \geqslant 0$, $\dot{h}_{si} + \rho_1 h_{si} \geq 0$ and the input constraint $w_i \in [-q_i\underline{g}_i, q_i\underline{g}_i]$ are satisfied simultaneously. When $x_{i2} > 0$, from (8), (9), and (11), it can be seen that $w_i = -q_i\underline{g}_i\mathrm{sgn}(x_{i2})$ is a feasible input, which satisfies all the constraints simultaneously. Similarly, when $x_{i2} < 0$, $w_i = -q_i\underline{g}_i\mathrm{sgn}(x_{i2})$ satisfies all the constraints (8), (9), and (12) simultaneously. Finally, when $x_{i2} = 0$, there only exists the input constraint (9), which means all $w_i \in [-q_i\underline{g}_i, q_i\underline{g}_i]$ are feasible. Therefore, under the conditions $h_{ri}^{+}(0) > 0$, $h_{ri}^{-}(0) > 0$, and $h_{si}(0) > 0$, there exists $w_i \in [-q_i\underline{g}_i, q_i\underline{g}_i]$ such that $h_{ri}^{+}(t) > 0$, $h_{ri}^{-}(t) > 0$ and $h_{si}(t) > 0$ hold for $\forall t \geq 0$. □

C. CBF-Based Safe Consensus Tracking

With the CLF-based consensus tracking condition and the CBF-based constraints handling conditions, a unified QP-based controller can be formulated to achieve safe consensus tracking of the multiagent system. Based on the two different CBF designs for the position constraints, two different control strategies are proposed.

First, we use the exponential CBF to handle the position constraint, which leads to the following QP problem for each agent:

$$\mathbf{QP1}: \min_{(w_i,\delta_i)} \delta_i^2 + w_i^2$$

$$\text{s.t.}\ \ 2(x_{i2} - x_{i2}^d)w_i + 2\alpha_i\left|x_{i2} - x_{i2}^d\right| + 2(x_{i2} - x_{i2}^d)(\hat{f}_i - \dot{x}_{i2}^d) + \lambda(x_{i2} - x_{i2}^d)^2 \leqslant \delta_i \tag{13}$$

$$- 2x_{i1}w_i - 2x_{i1}\hat{f}_i - 2\alpha_i\left|x_{i1}\right| - 2x_{i2}^2 - 2x_{i1}x_{i2}\left(\rho_2 + \rho_3\right) + \rho_2\rho_3\left(p_i^2 - x_{i1}^2\right) \geqslant 0 \tag{14}$$

$$- 2x_{i2}w_i - 2x_{i2}\hat{f}_i - 2\alpha_i\left|x_{i2}\right| + \rho_1\left(v_i^2 - x_{i2}^2\right) \geqslant 0 \tag{15}$$

$$|w_i| \leqslant \underline{g}_i q_i \tag{16}$$

where (13) is the relaxed CLF-based condition with the slack variable δ_i, (14) is the exponential CBF-based position constraint condition, (15) is the CBF-based velocity constraint condition, and (16) is the control input constraint. The performance index is to minimize the norm of the control input w_i and the violation of the CLF-based condition. The slack variable δ_i is introduced to put priority on guaranteeing safety of the agents. The following conclusion is obtained under the condition that there exists no conflict between the CBF-based conditions.

Theorem 3: Suppose that $h_{si}(0) > 0$, $\dot{h}_{ri}(0) + \rho_2 h_{ri}(0) > 0$, and $h_{ri}(0) > 0$. Under the condition that the QP 1 is feasible at each time instant, safe consensus tracking of (2) with guaranteed full state and input constraints is achieved with the control input $u_i = \frac{1}{g_i(x_i)}w_i^*$, where w_i^* is the optimal solution of QP 1.

Proof: When the QP 1 is feasible at each time instant, the CBF-based conditions (14) and (15) are satisfied along the trajectories of the agents. Therefore, under the initial conditions $h_{si}(0) > 0$, $\dot{h}_{ri}(0) + \rho_2 h_{ri}(0) > 0$, and $h_{ri}(0) > 0$, it follows that $h_{ri}(t) > 0$ and $h_{si}(t) > 0$ for all $t \geq 0$, which means safe consensus tracking with the full state and input constraints is achieved. □

However, even with the relaxed CLF-based condition, the QP 1 is not guaranteed to be feasible at all time since there may exist conflict between the position-based condition (14) and the velocity- and input-based conditions (15) and (16). To ensure the safety of the agents and the feasibility of the local QP problem, we propose the second control strategy based on the CBFs h_{pi}^{+} and h_{pi}^{-}

$$\mathbf{QP2}: \min_{(w_i,\delta_i)} \ \delta_i^2 + w_i^2$$

$$\text{s.t. } 2(x_{i2} - x_{i2}^d)w_i + 2\alpha_i \left|x_{i2} - x_{i2}^d\right|$$

$$+ 2(x_{i2} - x_{i2}^d)(\hat{f}_i - \dot{x}_{i2}^d) + \lambda(x_{i2} - x_{i2}^d)^2 \leqslant \delta_i \tag{17}$$

$$- \frac{x_{i2}}{\gamma_i}w_i - \frac{x_{i2}}{\gamma_i}\hat{f}_i - \alpha_i\frac{|x_{i2}|}{\gamma_i} - x_{i2}$$

$$+ \rho_4\left(p_i - x_{i1} - \frac{x_{i2}^2}{2\gamma_i}\right) \geqslant 0, \text{ if } x_{i2} > 0 \tag{18}$$

$$- \frac{x_{i2}}{\gamma_i}w_i - \frac{x_{i2}}{\gamma_i}\hat{f}_i - \alpha_i\frac{|x_{i2}|}{\gamma_i} + x_{i2}$$

$$+ \rho_4\left(p_i + x_{i1} - \frac{x_{i2}^2}{2\gamma_i}\right) \geqslant 0, \text{ if } x_{i2} < 0 \tag{19}$$

$$- 2x_{i2}w_i - 2x_{i2}\hat{f}_i - 2\alpha_i|x_{i2}| + \rho_1\left(v_i^2 - x_{i2}^2\right) \geqslant 0 \tag{20}$$

$$|w_i| \leqslant \underline{g}_i q_i. \tag{21}$$

Compared with QP 1, the position constraint condition is replaced by (18) and (19). From Theorem 2, we know that the QP 2 is always feasible, which leads to the following result.

Theorem 4: Under the conditions that $h_{ri}^{+}(0) > 0$, $h_{ri}^{-}(0) > 0$, and $h_{si}(0) > 0$, safe consensus tracking of (2) with guaranteed full state and input constraints is achieved with the control input $u_i = \frac{1}{g_i(x_i)}w_i^*$, where w_i^* is the optimal solution of QP 2. Moreover, for a static leader, which satisfies $|x_{01}| < p_i$, $i = 1, \ldots, N$, asymptotic consensus tracking is achieved while the full state and input constraints are guaranteed.

Proof: Note that with the control input u_i, all the CBF-based conditions are satisfied. Therefore, under the initial conditions $h_{ri}^{+}(0) > 0$, $h_{ri}^{-}(0) > 0$, and $h_{si}(0) > 0$, we have that $h_{ri}^{+}(t) > 0$, $h_{ri}^{-}(t) > 0$, and $h_{si}(t) > 0$ for all $t \geq 0$. It follows that safe consensus tracking of (2) with guaranteed full state and input constraints is achieved.

When x_{01} is a constant, we have $\dot{x}_{01} = 0$. Then, for the desired dynamics (3), we can set $k_2 = 0$ and the error dynamics (4) reduces to $\dot{\xi} = -k_1 H \tanh(\xi)$. It is easy to show that $\xi \to 0$ as $t \to \infty$. With the QP 2, the states of the agents will move toward the trajectory of ξ. Therefore, x_{i2} approaches zero as $t \to \infty$. Note that the CBF-based conditions are automatically satisfied as $x_{i2} \to 0$. Therefore, the optimal control input w_i^* will lead to $V_{if} \to 0$ and asymptotic consensus tracking is, therefore, achieved while the full state and input constraints are guaranteed.

Remark 4: Due to the presence of the unknown leader's information and the uncertain dynamics, robust relaxation terms are present in the CLF-CBF conditions of QP 1 and QP 2 such that the optimal solution w_i^* may not be Lipschitz continuous. When the leader's velocity and acceleration are available and the local dynamics $f_i(x_i)$ of the agents are known, the desired dynamics (3) can be taken as $\dot{x}_{i1} = -k_1\tanh(\xi_i) + \dot{x}_{i0}$, and the linear conditions in QP 1 and QP 2 can be rewritten with $\hat{f}_i(x_i) = f_i(x_i)$ and $\alpha_i = 0$. Then, following from the Mangasarian–Fromovitz regularity conditions [20], [25], the optimal control input is Lipschitz continuous in this case.

Remark 5: The proposed CBF-based distributed control strategy may also be extended to handle interagent constraints, such as the connectivity maintenance and collision avoidance constraints. By properly constructing the safety sets for connectivity and collision avoidance [13], similar CBF conditions can be derived and used to formulate the local QP problems to obtain the final control input for each agent. The main difficulty would be to define suitable safety sets such that the feasibility of the optimization problems can be guaranteed under both the local state and input constraints.

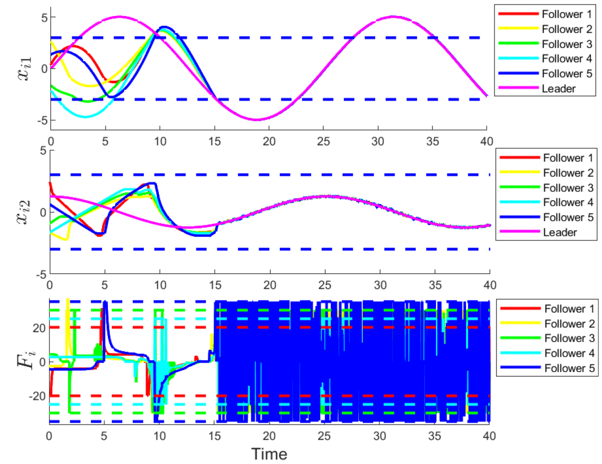

Fig. 1. States and inputs of the agents with the CLF-based consensus tracking controller.

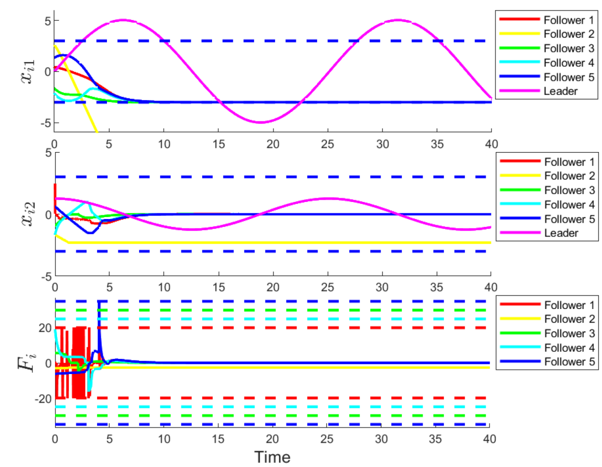

Fig. 2. States and inputs of the agents with the QP 1-based controller.

IV. Simulation

In this section, we consider a multirobot system with the second-order dynamics $\dot{r}_i = s_i$ and $m_i\dot{s}_i = F_i$, where $r_i = [r_{i1}, r_{i2}]^T \in \mathbb{R}^2$ is the position of robot i, $s_i = [s_{i1}, s_{i2}]^T \in \mathbb{R}^2$ is the velocity, $m_i \in \mathbb{R}$ is the mass, and $F_i = [F_{i1}, F_{i2}]^T \in \mathbb{R}^2$ is the control input. Suppose that only a nominal mass $\hat{m}_i$ is known for each robot, then the dynamics can be transformed into $\dot{r}_i = s_i$, $\dot{s}_i = \frac{1}{\hat{m}_i}F_i + \frac{\hat{m}_i - m_i}{m_i\hat{m}_i}F_i$.

Since the two dimensions of the position r_i are decoupled, we only illustrate the controller design for the first dimension. Denote $x_{i1} = r_{i1}$, $x_{i2} = s_{i1}$, $g_i = 1/\hat{m}_i$, $f_i = \frac{\hat{m}_i - m_i}{m_i\hat{m}_i}F_{i1}$, and $u_i = F_{i1}$, then we have the agent dynamics (2). Suppose that there are five followers. The trajectory of the leader is given by $x_{01}(t) = 5\sin(t/4)$. The interaction relation among the agents is determined by the matrix $H = [4, -1, 0, -1, -1;\ -1, 4, -1, -1, 0;\ 0, -1, 2, -1, 0;\ -1, -1, -1, 4, -1;\ -1, 0, 0, -1, 2]$. In the following, we will consider the effects of different position constraints. The velocity and input constraints of the agents are given by $[v_1, \ldots, v_5] = [3, 3, 3, 3, 3]$ and $[q_1, \ldots, q_5] = [20, 30, 30, 25, 35]$, respectively. The true and nominal masses of the robots are given by $[m_1, \ldots, m_5] = [5, 6, 7, 5, 8]$

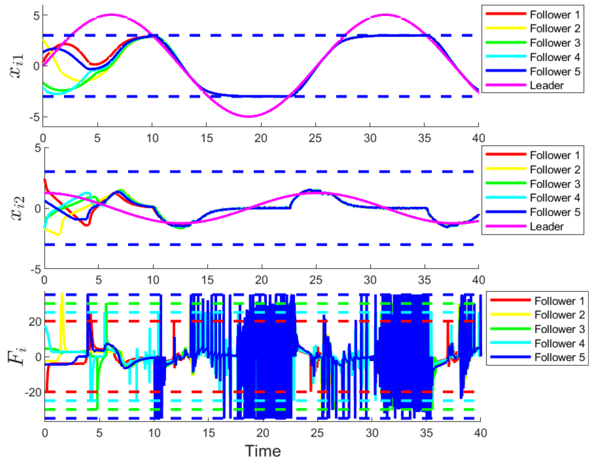

Fig. 3. States and inputs of the agents with the QP 2-based controller.

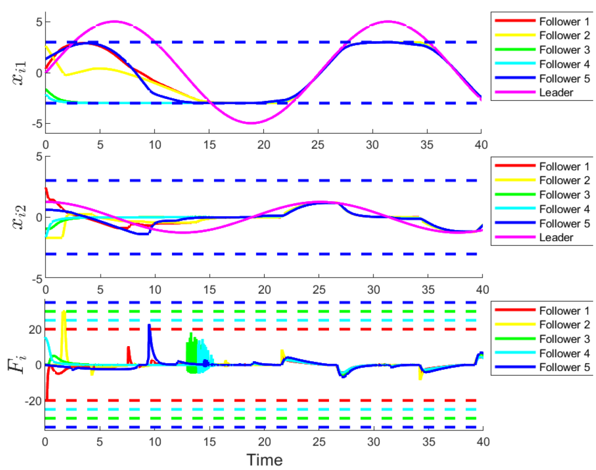

Fig. 4. States and inputs of the agents with the QP 2-based controller under known leader's information and local dynamics.

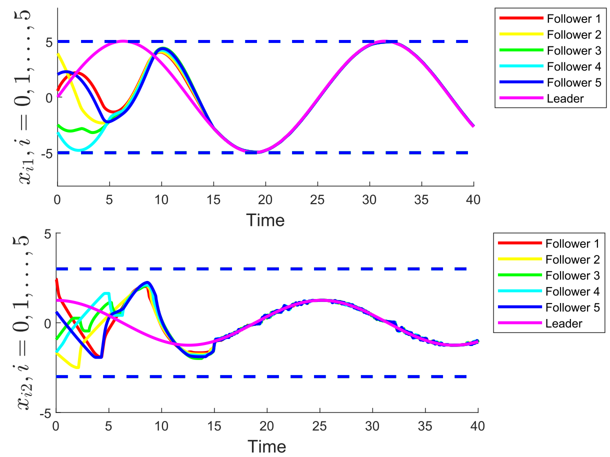

Fig. 5. Practical consensus tracking under the position constraints $[p_1, \ldots, p_5] = [5, 5, 5, 5, 5]$.

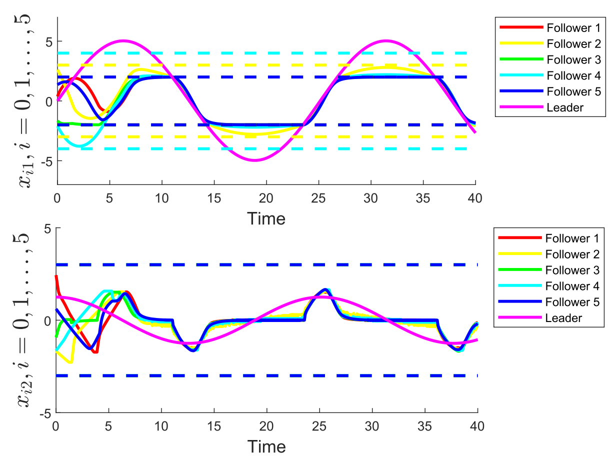

Fig. 6. Safe consensus tracking under the different position constraints $[p_1, \ldots, p_5] = [2, 3, 2, 4, 2]$.

and $[\hat{m}_1, \ldots, \hat{m}_5] = [4, 6.5, 6, 5.5, 9]$, respectively. By taking $\hat{f}_i = 0$, we have $[\alpha_1, \ldots, \alpha_5] = [1, 0.39, 0.72, 0.46, 0.49]$ and $\beta_i = 0$, $i = 1, \ldots, 5$. For the desired dynamics (3), the controller parameters are chosen as $k_1 = 2$, $k_2 = 1.5$, and $\varepsilon = 0.01$. For the CLF-based conditions (13) and (17), $\lambda = 0.1$. For the CBF-based conditions (14), (15), and (18)–(20), $\rho_1 = \rho_2 = \rho_3 = \rho_4 = 1$.

First, we consider the position constraint $[p_1, \ldots, p_5] = [3, 3, 3, 3, 3]$. Note that under these position constraints, the robots cannot track the leader's trajectory outside the region $[-3, 3]$. To show the effectiveness of the CLF-based consensus tracking method and the necessity of the CBF-based constraints, we first consider the QP-based control input where only the CLF-based condition is imposed, i.e., we solve the QP 1 with only the constraints (13) and (16) or the QP 2 with only the constraints (17) and (21). The simulation results are shown in Fig. 1. It can be seen that practical consensus tracking is successfully achieved with the CLF-based controller. However, the position constraints of the agents are violated during the tracking process. Then, we consider the case with the proposed CBF-based controller. For the position constraint, the high-order CBF is first employed, i.e., QP 1 is used to obtain the control input. The simulation results are shown in Fig. 2. It can be seen that due to the conflict between the position, velocity, and input constraints, safe consensus tracking cannot be achieved with the QP 1. Then, the position constraint is handled by the proposed CBFs h^+_{pi} and h^-_{pi}, and QP 2 is used to produce the control input. The simulation results are shown in Fig. 3. It can be seen that safe consensus tracking is achieved while the position, velocity, and input constraints are guaranteed. The agents cooperatively track the trajectory of the leader only when the leader's trajectory lies in the position constraints of all the robots. On the simulation machine with Intel Core i7-6700 CPU@3.40 GHz and 8 GB RAM, the maximum solving time of the local QP problems for each agent is under 0.05 s, which shows the practical application potential of the proposed control strategy.

Note that in Fig. 3, there exists chattering phenomenon due to the unknown leader's information and the uncertain dynamics. When the leader's information is available and the dynamics of the agents are known, the simulation results are shown in Fig. 4, which shows that safe consensus tracking with continuous control inputs is achieved in this case. Next, we consider the position constraints $[p_1, \ldots, p_5] = [5, 5, 5, 5, 5]$ and $[p_1, \ldots, p_5] = [2, 3, 2, 4, 2]$, respectively. In the former case, the desired trajectory lies in the position constraints of the robots, whereas in the latter case, not all the agents could track the desired trajectory without violating the position constraint. With the QP 2-based controllers, the position tracking results are shown in Figs. 5 and 6, respectively. It verifies that in both cases, safe consensus tracking is achieved while all the position, velocity, and input constraints are satisfied.

V. Conclusion

In this work, we have considered the safe consensus tracking problem for second-order uncertain nonlinear multiagent systems subject to both state and input constraints. The leader's trajectory is not required to satisfy all the followers' state and input constraints. By constructing the CLF-based consensus tracking condition and the CBF-based state and input constraints conditions, two QP-based controllers have been proposed to achieve the safe consensus tracking objective. It was proven that with one of the QP-based controller, consensus tracing with guaranteed state and input constraints can be ensured. Simulation examples verify the effectiveness of the proposed control strategy. Future work includes considering collision avoidance with respect to obstacles in the environment and handling more general cooperative tasks.

References

[1] J. Cortes and M. Egerstedt, "Coordinated control of multi-robot systems: A survey," *SICE J. Control, Meas., Syst. Integration*, vol. 10, no. 6, pp. 495–503, 2017.

[2] Y. Cao, W. Yu, W. Ren, and G. Chen, "An overview of recent progress in the study of distributed multi-agent coordination," *IEEE Trans. Ind. Informat.*, vol. 9, no. 1, pp. 427–438, Feb. 2013.

[3] J. Qin, Q. Ma, Y. Shi, and L. Wang, "Recent advances in consensus of multi-agent systems: A brief survey," *IEEE Trans. Ind. Electron.*, vol. 64, no. 6, pp. 4972–4983, Jun. 2017.

[4] K. K. Oh, M. C. Park, and H. S. Ahn, "A survey of multi-agent formation control," *Automatica*, vol. 53, pp. 424–440, 2015.

[5] Y. Li, J. Xiang, and W. Wei, "Consensus problems for linear time-invariant multi-agent systems with saturation constraints," *IET Control Theory Appl.*, vol. 5, no. 6, pp. 823–829, May 2011.

[6] W. Ren, "On consensus algorithms for double-integrator dynamics," *IEEE Trans. Autom. Control*, vol. 53, no. 6, pp. 1503–1509, Jul. 2008.

[7] Z. Zhao and Z. Lin, "Global leader-following consensus of a group of general linear systems using bounded controls," *Automatica*, vol. 68, pp. 294–304, 2016.

[8] H. Su, M. Z. Q. Chen, J. Lam, and Z. Lin, "Semi-global leader-following consensus of linear multi-agent systems with input saturation via low gain feedback," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 60, no. 7, pp. 1881–1889, Jul. 2013.

[9] G. Ferrari-Trecate, L. Galbusera, M. P. E. Marciandi, and R. Scattolini, "Model predictive control schemes for consensus in multi-agent systems with single- and double-integrator dynamics," *IEEE Trans. Autom. Control*, vol. 54, no. 11, pp. 2560–2572, Nov. 2009.

[10] J. Zhan and X. Li, "Flocking of multi-agent systems via model predictive control based on position-only measurements," *IEEE Trans. Ind. Informat.*, vol. 9, no. 1, pp. 377–385, Feb. 2013.

[11] P. Lin, W. Ren, C. Yang, and W. Gui, "Distributed consensus of second-order multiagent systems with nonconvex velocity and control input constraints," *IEEE Trans. Autom. Control*, vol. 63, no. 4, pp. 1171–1176, Apr. 2018.

[12] J. Fu, G. Wen, W. Yu, T. Huang, and X. Yu, "Consensus of second-order multi-agent systems with both velocity and input constraints," *IEEE Trans. Ind. Electron.*, vol. 66, no. 10, pp. 7946–7955, Oct. 2019.

[13] J. Fu, Y. Lv, G. Wen, X. Yu, and T. Huang, "Velocity and input constrained coordination of second-order multi-agent systems with relative output information," *IEEE Trans. Netw. Syst. Eng.*, vol. 7, no. 3, pp. 1925–1938, Jul./Sep. 2020.

[14] Y. Zhang, H. Liang, H. Ma, Q. Zhou, and Z. Yu, "Distributed adaptive consensus tracking control for nonlinear multi-agent systems with state constraints," *Appl. Math. Comput.*, vol. 326, pp. 16–32, 2018.

[15] F. Yuan, Y. Ma, Y. J. Liu, J. Lan, and T. Xu, "Adaptive distributed tracking control for non-affine multi-agent systems with state constraints and dead-zone input," *J. Franklin Inst.*, vol. 359, no. 1, pp. 352–370, 2022.

[16] T. Zhang, M. Xia, and Y. Yi, "Adaptive neural dynamic surface control of strict-feedback nonlinear systems with full state constraints and unmodeled dynamics," *Automatica*, vol. 81, pp. 232–239, 2017.

[17] Y. Liu, H. Zhang, J. Sun, and Y. Wang, "Adaptive fuzzy containment control for multiagent systems with state constraints using unified transformation functions," *IEEE Trans. Fuzzy Syst.*, vol. 30, no. 1, pp. 162–174, Jan. 2022.

[18] F. Zhang, G. Tan, C. Yu, N. Ding, C. Song, and M. Liu, "Fair transmission rate adjustment in cooperative vehicle safety systems based on multi-agent model predictive control," *IEEE Trans. Veh. Technol.*, vol. 66, no. 7, pp. 6115–6129, Jul. 2017.

[19] H. Li and Y. Shi, "Robust distributed model predictive control of constrained continuous-time nonlinear systems: A robustness constraint approach," *IEEE Trans. Autom. Control*, vol. 59, no. 6, pp. 1673–1678, Jun. 2014.

[20] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs for safety critical systems," *IEEE Trans. Autom. Control*, vol. 62, no. 8, pp. 3861–3876, Aug. 2017.

[21] R. Grandia, A. J. Taylor, A. Singletary, M. Hutter, and A. D. Ames, "Nonlinear model predictive control of robotic systems with control Lyapunov functions," 2020, *arXiv:2006.01229*.

[22] P. Glotfelter, I. Buckley, and M. Egerstedt, "Hybrid nonsmooth barrier functions with applications to provably safe and composable collision avoidance for robotic systems," *IEEE Robot. Autom. Lett.*, vol. 4, no. 2, pp. 1303–1310, Apr. 2019.

[23] M. M. Polycarpou, "Stable adaptive neural control scheme for nonlinear systems," *IEEE Trans. Autom. Control*, vol. 41, no. 3, pp. 447–451, Mar. 1996.

[24] Q. Nguyen and K. Sreenath, "Exponential control barrier functions for enforcing high relative-degree safety-critical constraints," in *Proc. Amer. Control Conf.*, 2016, pp. 322–328.

[25] B. Morris, M. J. Powell, and A. D. Ames, "Sufficient conditions for the Lipschitz continuity of QP-based multi-objective control of humanoid robots," in *Proc. IEEE 52nd Conf. Decis. Control*, 2013, pp. 2920–2926.